

# Supplementary: Effects of Explanations on Driver Situation Awareness Under Physical Adversarial Attacks in Automated Driving: A Driving Simulator Study

## Overview

This supplementary document accompanies the paper *Effects of Explanations on Driver Situation Awareness Under Physical Adversarial Attacks in Automated Driving: A Driving Simulator Study*. It contains the complete study instruments and statistics results.

## A Instruments

### A.1 SAGAT Questions

#### Questions After Each STOP sign Attack Scenario

Please answer the following questions based on your recall of the situation where you take over the control.

- (1) The simulation was just frozen. Which object was in front of the AV before your takeover control?
  - a. Vehicle
  - b. Pedestrian
  - c. Traffic Sign
  - d. I don't know
  - e. Other \_\_\_\_\_
- (2) What caused the takeover control in that situation?
  - a. Pedestrian's intention to cross the street
  - b. Approaching traffic sign
  - c. Faulty road lanes
  - d. I don't know
  - e. Other \_\_\_\_\_
- (3) If there was no takeover control, what situation would be?
  - a. Other road user's violations
  - b. AV's possibility to hit pedestrian
  - c. Violating the traffic rules
  - d. I don't know
  - e. Other \_\_\_\_\_

#### Questions After Each Pedestrian Attack Scenario

Please answer the following questions based on your recall of the situation where you take over the control.

- (1) The simulation was just frozen. Which object was in front of the AV before your takeover control?
  - a. Vehicle
  - b. Pedestrian
  - c. Traffic Sign
  - d. I don't know
  - e. Other \_\_\_\_\_
- (2) What caused the takeover control in that situation?
  - a. Pedestrian's intention to cross the street
  - b. Approaching traffic sign
  - c. Faulty road lanes
  - d. I don't know
  - e. Other \_\_\_\_\_
- (3) If there was no takeover control, what situation would be?
  - a. Impeding the traffic by stopping at the roadside

- b. Other road user's violations
- c. AV's possibility to hit pedestrian
- d. I don't know
- e. Other \_\_\_\_\_

#### Questions After Each Vehicle Attack Scenario

Please answer the following questions based on your recall of the situation where you take over the control.

- (1) The simulation was just frozen. Which object was in front of the AV before your takeover control?
  - a. Vehicle
  - b. Pedestrian
  - c. Traffic Sign
  - d. I don't know
  - e. Other \_\_\_\_\_
- (2) What caused the takeover control in that situation?
  - a. Approaching traffic sign
  - b. Faulty road lanes
  - c. Weird vehicle body trim
  - d. I don't know
  - e. Other \_\_\_\_\_
- (3) If there was no takeover control, what situation would be?
  - a. Other road user's violations
  - b. Possibility of rear-end collision
  - c. Impeding the traffic by stopping at the roadside
  - d. I don't know
  - e. Other \_\_\_\_\_

#### Questions After Each Road patch Attack Scenario

Please answer the following questions based on your recall of the situation where you take over the control.

- (1) The simulation was just frozen. Which object was in front of the AV before your takeover control?
  - a. Vehicle
  - b. Abnormal road surface
  - c. Traffic Sign
  - d. I don't know
  - e. Other \_\_\_\_\_
- (2) What caused the takeover control in that situation?
  - a. Approaching traffic sign
  - b. Dirty road surface
  - c. Weird vehicle body trim
  - d. I don't know
  - e. Other \_\_\_\_\_
- (3) If there was no takeover control, what situation would be?
  - a. Other road user's violations
  - b. Possibility of changing lane
  - c. Violating the traffic rules
  - d. I don't know
  - e. Other \_\_\_\_\_

**Questions for Baseline Situation Awareness (SA) Test** (Used in practice drive. It controls for individual differences in initial SA.) Please answer the following questions based on your recall of the situation where you take over the control.

- (1) The simulation was just frozen. Which object was in front of the AV when you take over?
  - a. Vehicle
  - b. Pedestrian
  - c. Traffic Light
  - d. I don't know
  - e. Other \_\_\_\_\_
- (2) What caused the takeover control in that situation?
  - a. Pedestrian's intention to cross the street
  - b. Approaching traffic sign
  - c. Approaching traffic light
  - d. I don't know
  - e. Other \_\_\_\_\_
- (3) If there was no takeover control, what situation would be?
  - a. Other road user's violations
  - b. AV's possibility to hit pedestrian
  - c. Violating traffic rules
  - d. I don't know
  - e. Other \_\_\_\_\_

## A.2 Trust Evaluation

Based on the drive you just completed, please answer on a 7-point scale (1 = not at all, 7 = extremely).

- (1) I trust the automation in this situation.
- (2) I would have performed better than the automation in this situation. (Reverse scored.)
- (3) In this situation, the automated vehicle performs well enough for me to engage in other activities (such as reading).
- (4) The situation was risky. (Reverse scored.)
- (5) The automated vehicle made an unsafe judgment in this situation. (Reverse scored.)
- (6) The automated vehicle reacted appropriately to the environment.

## A.3 Mental Workload Evaluation

(A revised version of NASA-TLX.)

Based on the drive you just completed, please answer on a 7-point scale (1 = not at all, 7 = extremely).

- (1) I was mentally taxed when perceiving the situation through the display.
- (2) I had to concentrate a lot to understand the display.
- (3) I found it difficult to understand the situation through the display.
- (4) I quickly understood the situation through the display. (Reverse scored.)
- (5) I felt relaxed when perceiving the situation through the display. (Reverse scored.)
- (6) I felt unsatisfied with how the display helped me understand the situation.

## A.4 Post-study Open-ended Questions

- (1) Using your own words, please describe how the **warnings** influenced your takeover control performance.
- (2) The following screenshots (see Figure 1) show the four different scenarios we presented in the simulated driving. Please recall your experience for each scenario and answer the questions below.



**Figure 1: Screenshots shown to participants for the open-ended recognition questions.**

**Please describe any abnormality you detected in each scenario.**

- a. Top left scenario:
- b. Top right scenario:
- c. Bottom left scenario:
- d. Bottom right scenario:
- (3) Please describe anything in the simulated driving that helped you better understand each scenario.
- (4) Do you have any suggestions to better inform human drivers of such attacks?

## B Statistic Results

### B.1 Descriptive statistics

**Table 1: Overall Driver SA Results. Participant’s baseline SA (*Initial\_SA*) is reported by group (with vs. without explanation). *SA\_Change* is reported by attack object, group, system reliability, and trial order.**

| Variable                                 | Condition                   | N  | Mean  | SE    |
|------------------------------------------|-----------------------------|----|-------|-------|
| <i>Initial_SA</i> (by group)             | Without Explanation         | 16 | 2.81  | 0.188 |
|                                          | With Explanation            | 16 | 2.25  | 0.266 |
| <i>SA_change</i> (by attack object)      | Stop Sign                   | 32 | 0.19  | 0.165 |
|                                          | Vehicle                     | 32 | 0.19  | 0.165 |
|                                          | Pedestrian                  | 32 | 0.16  | 0.196 |
|                                          | Lane Markings               | 32 | -0.19 | 0.239 |
| <i>SA_change</i>                         | <i>Without Explanation:</i> |    |       |       |
|                                          | Stop Sign                   | 16 | -0.12 | 0.125 |
|                                          | Vehicle                     | 16 | -0.12 | 0.125 |
|                                          | Pedestrian                  | 16 | -0.12 | 0.202 |
|                                          | Lane Markings               | 16 | -0.69 | 0.254 |
|                                          | <i>With Explanation:</i>    |    |       |       |
|                                          | Stop Sign                   | 16 | 0.50  | 0.289 |
|                                          | Vehicle                     | 16 | 0.50  | 0.289 |
|                                          | Pedestrian                  | 16 | 0.44  | 0.329 |
|                                          | Lane Markings               | 16 | 0.31  | 0.373 |
| <i>SA_change</i> (by system reliability) | Low                         | 32 | 0.06  | 0.179 |
|                                          | High                        | 32 | 0.16  | 0.163 |
| <i>SA_change</i>                         | <i>Without Explanation:</i> |    |       |       |
|                                          | Low                         | 16 | -0.29 | 0.155 |
|                                          | High                        | 16 | -0.19 | 0.101 |
|                                          | <i>With Explanation:</i>    |    |       |       |
|                                          | Low                         | 16 | 0.42  | 0.304 |
|                                          | High                        | 16 | 0.50  | 0.289 |
| <i>SA_change</i> (by trial order)        | 1st                         | 32 | -0.03 | 0.208 |
|                                          | 2nd                         | 32 | 0.12  | 0.194 |
|                                          | 3rd                         | 32 | 0.03  | 0.203 |
|                                          | 4th                         | 32 | 0.22  | 0.172 |

**Table 2: Participant’s *SA\_Change* across the three SA levels, separated by group (with vs. without explanation).**

| Group               | SA Level | Mean   | SE    |
|---------------------|----------|--------|-------|
| Without Explanation | Level 1  | -0.031 | 0.050 |
|                     | Level 2  | -0.109 | 0.045 |
|                     | Level 3  | -0.125 | 0.040 |
| With Explanation    | Level 1  | 0.078  | 0.093 |
|                     | Level 2  | 0.203  | 0.115 |
|                     | Level 3  | 0.156  | 0.120 |

**Table 3: Takeover Performance Results (Takeover response time for longitudinal and lateral controls). Data is reported by attack object, system reliability, and their respective interactions with the group (with vs. without explanation). Note:  $RT_{long}$  = response time for longitudinal control;  $RT_{lat}$  = response time for lateral control.**

| Variable                          | Condition                   | N  | $RT_{long}$ |       | $RT_{lat}$ |       |
|-----------------------------------|-----------------------------|----|-------------|-------|------------|-------|
|                                   |                             |    | Mean        | SE    | Mean       | SE    |
| <i>RT</i> (by attack object)      | Stop Sign                   | 32 | 2.88        | 0.175 | 5.10       | 0.481 |
|                                   | Vehicle                     | 32 | 2.65        | 0.236 | 5.53       | 0.588 |
|                                   | Pedestrian                  | 32 | 2.64        | 0.124 | 6.52       | 0.568 |
|                                   | Lane Markings               | 32 | 3.26        | 0.272 | 5.85       | 0.385 |
| <i>RT</i>                         | <i>Without Explanation:</i> |    |             |       |            |       |
|                                   | Stop Sign                   | 16 | 2.60        | 0.226 | 4.54       | 0.447 |
|                                   | Vehicle                     | 16 | 2.38        | 0.271 | 5.92       | 1.049 |
|                                   | Pedestrian                  | 16 | 2.52        | 0.182 | 6.64       | 0.917 |
|                                   | Lane Markings               | 16 | 2.93        | 0.367 | 5.72       | 0.654 |
|                                   | <i>With Explanation:</i>    |    |             |       |            |       |
|                                   | Stop Sign                   | 16 | 3.17        | 0.255 | 5.66       | 0.846 |
|                                   | Vehicle                     | 16 | 2.92        | 0.383 | 5.14       | 0.554 |
|                                   | Pedestrian                  | 16 | 2.76        | 0.168 | 6.40       | 0.701 |
|                                   | Lane Markings               | 16 | 3.60        | 0.396 | 5.99       | 0.427 |
| <i>RT</i> (by system reliability) | Low                         | 32 | 2.82        | 0.172 | 5.78       | 0.323 |
|                                   | High                        | 32 | 2.97        | 0.197 | 5.68       | 0.545 |
| <i>RT</i>                         | <i>Without Explanation:</i> |    |             |       |            |       |
|                                   | Low                         | 16 | 2.59        | 0.218 | 5.52       | 0.515 |
|                                   | High                        | 16 | 2.65        | 0.276 | 6.26       | 0.966 |
|                                   | <i>With Explanation:</i>    |    |             |       |            |       |
|                                   | Low                         | 16 | 3.05        | 0.260 | 6.03       | 0.396 |
|                                   | High                        | 16 | 3.28        | 0.265 | 5.09       | 0.500 |

**Table 4: Trust and Mental Workload Results. Data is reported by group, system reliability, and their respective interactions.**

| Variable                              | Condition                   | N  | Trust |       | Mental Workload |       |
|---------------------------------------|-----------------------------|----|-------|-------|-----------------|-------|
|                                       |                             |    | Mean  | SE    | Mean            | SE    |
| <i>Scores</i> (by group)              | Without Explanation         | 16 | 4.57  | 0.197 | 2.45            | 0.209 |
|                                       | With Explanation            | 16 | 4.75  | 0.197 | 2.09            | 0.209 |
| <i>Scores</i> (by system reliability) | Low                         | 32 | 4.59  | 0.164 | 2.21            | 0.136 |
|                                       | High                        | 32 | 4.73  | 0.160 | 2.33            | 0.177 |
| <i>Scores</i>                         | <i>Without Explanation:</i> |    |       |       |                 |       |
|                                       | Low                         | 16 | 4.47  | 0.295 | 2.30            | 0.211 |
|                                       | High                        | 16 | 4.67  | 0.272 | 2.60            | 0.255 |
|                                       | <i>With Explanation:</i>    |    |       |       |                 |       |
|                                       | Low                         | 16 | 4.71  | 0.146 | 2.12            | 0.173 |
|                                       | High                        | 16 | 4.79  | 0.168 | 2.06            | 0.245 |

## B.2 Inferential statistic

**Table 5: Analyses of Variance for Situation Awareness Change (SA\_change) and Takeover Response Time for Longitudinal Control ( $RT_{long}$ ) and Lateral Control ( $RT_{lat}$ ) as a Function of Attack Type (STOP sign, Pedestrian, Vehicle, and Road Lane Marking) as Within-Subjects Factor and Explanation (with vs. without) as a Between-Subjects Factor. Note: df= degrees of freedom; RT= response time; Because the sphericity assumption was violated for terms involving the attack type factor (epsilons less than .75), the reported p-values are adjusted by Greenhouse–Geisser correction [1]. Bold font indicates statistical significance ( $p \leq .05$ ).**

| DV          | Effect                      | df          | <i>F</i>    | <i>p</i>    | $\eta_p^2$  |
|-------------|-----------------------------|-------------|-------------|-------------|-------------|
| SA_change   | Explanation                 | 1, 30       | <b>4.97</b> | <b>.033</b> | <b>.142</b> |
|             | Attack                      | 2.36, 70.73 | 2.63        | .070        | .081        |
|             | Explanation $\times$ Attack | 2.36, 70.73 | 0.79        | .477        | .026        |
| $RT_{long}$ | Explanation                 | 1, 30       | 2.32        | .138        | .072        |
|             | Attack                      | 2.44, 73.34 | <b>4.10</b> | <b>.014</b> | <b>.120</b> |
|             | Explanation $\times$ Attack | 2.44, 73.34 | 0.42        | .701        | .014        |
| $RT_{lat}$  | Explanation                 | 1, 30       | 0.02        | .901        | < .001      |
|             | Attack                      | 2.69, 80.64 | 1.94        | .136        | .061        |
|             | Explanation $\times$ Attack | 2.69, 80.64 | 0.88        | .444        | .029        |

**Table 6: Tests for Reliability  $\times$  Explanation (between: with vs. without). Effect sizes are partial eta squared ( $\eta_p^2$ ).**

| DV          | Effect                           | df    | <i>F</i>    | <i>p</i>    | $\eta_p^2$  |
|-------------|----------------------------------|-------|-------------|-------------|-------------|
| SA_change   | Explanation                      | 1, 30 | <b>5.40</b> | <b>.027</b> | <b>.153</b> |
|             | Reliability                      | 1, 30 | 0.60        | .444        | .020        |
|             | Explanation $\times$ Reliability | 1, 30 | 0.01        | .932        | <.001       |
| $RT_{long}$ | Explanation                      | 1, 30 | 2.65        | .114        | .081        |
|             | Reliability                      | 1, 30 | 1.12        | .299        | .036        |
|             | Explanation $\times$ Reliability | 1, 30 | 0.40        | .534        | .013        |
| $RT_{lat}$  | Explanation                      | 1, 30 | 0.16        | .689        | .005        |
|             | Reliability                      | 1, 30 | 0.07        | .792        | .002        |
|             | Explanation $\times$ Reliability | 1, 30 | <b>5.23</b> | <b>.029</b> | <b>.148</b> |

**Table 7: Tests for Reliability  $\times$  Explanation on secondary outcomes (Trust, Mental\_workload). Partial eta squared ( $\eta_p^2$ ) reported.**

| DV              | Effect                           | df    | <i>F</i> | <i>p</i> | $\eta_p^2$ |
|-----------------|----------------------------------|-------|----------|----------|------------|
| Trust           | Explanation                      | 1, 30 | 0.42     | .520     | .014       |
|                 | Reliability                      | 1, 30 | 0.73     | .401     | .024       |
|                 | Explanation $\times$ Reliability | 1, 30 | 0.12     | .730     | .004       |
| Mental_workload | Explanation                      | 1, 30 | 1.53     | .225     | .049       |
|                 | Reliability                      | 1, 30 | 1.24     | .274     | .040       |
|                 | Explanation $\times$ Reliability | 1, 30 | 2.51     | .124     | .077       |

## References

- [1] Ellen R Girden. 1992. *ANOVA: Repeated measures*. Number 84. Sage.

## C Demographics

**Table 8: Demographics of participants in the study.**

| Item                                  | Category                     | Count | %     |
|---------------------------------------|------------------------------|-------|-------|
| Gender                                | Female                       | 11    | 34.4% |
|                                       | Male                         | 21    | 65.6% |
| Age                                   | 18–24                        | 20    | 62.5% |
|                                       | 25–34                        | 7     | 21.9% |
|                                       | 35–44                        | 3     | 9.4%  |
|                                       | 45–55                        | 1     | 3.1%  |
|                                       | >55                          | 1     | 3.1%  |
| Ethnicity                             | White                        | 18    | 56.3% |
|                                       | Asian                        | 13    | 40.6% |
|                                       | Black/African American       | 1     | 3.1%  |
| Education                             | High school graduate         | 1     | 3.1%  |
|                                       | Some college                 | 16    | 50.0% |
|                                       | Bachelor degree              | 5     | 15.6% |
|                                       | Master degree                | 5     | 15.6% |
|                                       | Doctoral degree              | 3     | 9.4%  |
|                                       | Others                       | 2     | 6.3%  |
| Years of Driving                      | Less than 1 year             | 3     | 9.4%  |
|                                       | 1–5 years                    | 19    | 59.4% |
|                                       | 6–10 years                   | 7     | 21.9% |
|                                       | More than 10 years           | 3     | 9.4%  |
| Driving Frequency                     | Everyday or nearly every day | 13    | 40.6% |
|                                       | Once or twice a week         | 12    | 37.5% |
|                                       | Once or twice a month        | 3     | 9.4%  |
|                                       | A few times a year           | 4     | 12.5% |
| Driving Assistance-Related Experience | Yes                          | 11    | 34.4% |
|                                       | No                           | 21    | 65.6% |
| Automated Driving-Related Experience  | Yes                          | 3     | 9.4%  |
|                                       | No                           | 29    | 90.6% |

## D Codebook

| Question | Theme                                 | Code                                       | Count | Include when participant says...                                                                                            |
|----------|---------------------------------------|--------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------|
| Q1       | Perceived warning effectiveness       | Effectiveness                              | 20    | The warning helped them react, stop another task, brake, or generally improve performance.                                  |
|          | Guided timely takeover action         | Takeover trigger                           | 16    | Participant links the warning to taking over, braking, steering, or resuming control.                                       |
|          | Mixed but generally positive attitude | Attitude                                   | 9     | Participant shows an attitude or has positive judgment.                                                                     |
|          | Modality                              | Modality                                   | 7     | Participant mentions beep, sound, audio, or visual warning.                                                                 |
|          | Attention                             | Color                                      | 4     | Participant mentions red color or color salience.                                                                           |
|          |                                       | Attention                                  | 6     | Participant says the warning got their attention, made them focus, or pulled them back to the road.                         |
|          | Situation comprehension               | Comprehension                              | 4     | Participant mentions knowing the problem, understanding the situation, or interpreting system trouble.                      |
|          | Preparation                           | Preparedness                               | 2     | Participant says the warning prepared them, made them ready, or moved feet/hands into position.                             |
| Q2       | Attack interpretation / specificity   | Other suggestions                          | 3     | Participant proposes timing or other improvement.                                                                           |
|          |                                       | Abnormality                                | 71    | Participant describes something unusual, confusing, wrong, or abnormal in the scene/system.                                 |
|          |                                       | Recognize attack                           | 15    | Participant clearly interprets the event as an attack.                                                                      |
|          | Unusable response                     | Did not recognize anything                 | 34    | Participant says nothing related to attack object.                                                                          |
| Q3       | Explanation                           | N/A                                        | 8     | Response is not applicable, or unavailable.                                                                                 |
|          | Explanation                           | Explanations                               | 9     | Participant says explanation/yellow warning/text helped them understand the situation.                                      |
|          | Explanation + Warning                 | Both warning and explanations              | 2     | Participant links both warning and explanatory information to scenario understanding.                                       |
|          | Warning                               | Warning                                    | 7     | Participant says the warning from the AV system helped them know the hazard/problem.                                        |
|          | Auditory cue                          | Beep sound                                 | 5     | Participant identifies sound/beep/alarm as aiding awareness or understanding.                                               |
|          | No reported aid                       | None                                       | 7     | Response is a substantive no-help answer.                                                                                   |
|          | Text/visual cue                       | Text                                       | 2     | Participant emphasizes visual textual cue but not necessarily explanatory reasoning.                                        |
|          | Interface display cue                 | Head up display                            | 1     | Participant mentions HUD or heads-up display.                                                                               |
|          | Practice                              | Practice session                           | 1     | Participant names the practice session as the aid.                                                                          |
| Q4       | Drivers demand                        | Require more attention than manual driving | 1     | Participant contrasts AV monitoring demands with manual driving.                                                            |
|          | Increasing driver awareness           | Increase awareness                         | 6     | Participant recommends awareness, outreach, media, or general public communication.                                         |
|          |                                       | Education and training                     | 3     | Participant recommends education, training, outreach, or learning what AVs struggle to detect.                              |
|          | Improved warning design               | Warning improvement                        | 5     | Participant proposes earlier warnings, distance-to-hazard, bigger text, louder alert, personalization, or highlight source. |
|          |                                       | Multimodal cues                            | 4     | Participant proposes multimodal warning designs (e.g., integrating haptic and vibration signals).                           |
|          | Other suggestions                     | Other suggestions                          | 5     | Participant proposes a recommendation outside awareness/training/warning modality or timing.                                |
|          | Driving behavior / human attention    | Driving behavior / human attention         | 2     | Participant recommends keeping attention on the road or avoiding distractions.                                              |
|          | None                                  | None                                       | 11    | No suggestion.                                                                                                              |